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Division: A2

ASSIGNMENT No: 11

TITLE: Write a Java program to demonstrate the implementation of interfaces.

Problem Statement: Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives: To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory: An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Exercise:

1. Create an interface Printable and implement it in Student and Employee classes.

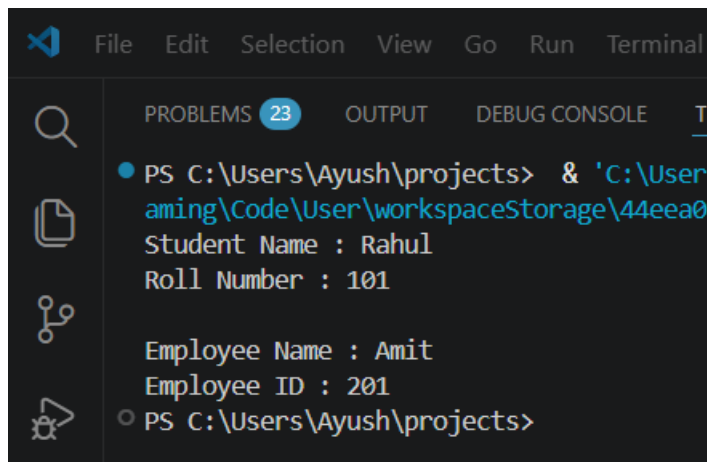
Program:

```
File Edit Selection View Go Run Terminal Help

J Main12.java 2 X
helloworld > J Main12.java > Language Support for Java(TM) by Red Hat > Employee

1  interface Printable
2  {
3      void print();
4  }
5
6  class Stude implements Printable
7  {
8      String name;
9      int rollNo;
10
11     Stude(String name, int rollNo)
12     {
13         this.name = name;
14         this.rollNo = rollNo;
15     }
16
17     public void print()
18     {
19         System.out.println("Student Name : " + name);
20         System.out.println("Roll Number : " + rollNo);
21     }
22 }
23
24 class Employee implements Printable
25 {
26     String name;
27     int empId;
28
29     Employee(String name, int empId)
30     {
31         this.name = name;
32         this.empId = empId;
33     }
34
35     public void print()
36     {
37         System.out.println("Employee Name : " + name);
38         System.out.println("Employee ID : " + empId);
39     }
40 }
41
42 public class Main12
43 {
44     Run main | Debug main | Run | Debug
45     public static void main(String[] args)
46     {
47         Stude s = new Stude(name: "Rahul", rollNo: 101);
48         Employee e = new Employee(name: "Amit", empId: 201);
49
50         s.print();
51         System.out.println();
52         e.print();
53     }
54 }
```

Output:

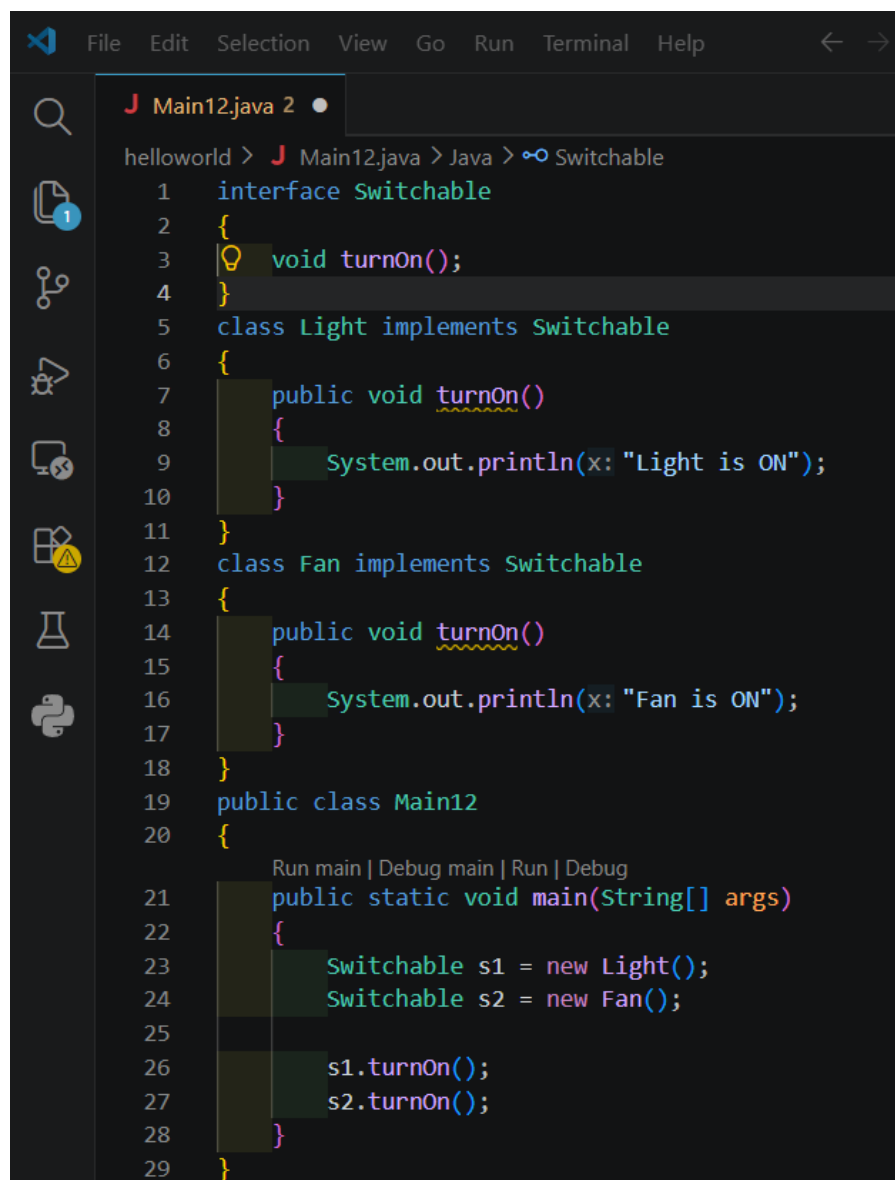


```
PS C:\Users\Ayush\projects> & 'C:\User
aming\Code\User\workspaceStorage\44eea0
Student Name : Rahul
Roll Number : 101

Employee Name : Amit
Employee ID : 201
PS C:\Users\Ayush\projects>
```

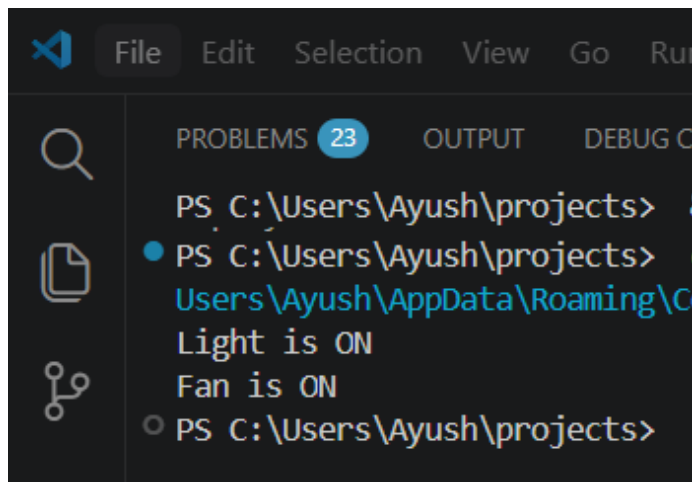
2. Create an interface Switchable with a method turnOn(). Implement the interface in Light and Fan classes to display the device status.

Program:



```
1 interface Switchable
2 {
3     void turnOn();
4 }
5 class Light implements Switchable
6 {
7     public void turnOn()
8     {
9         System.out.println(x: "Light is ON");
10    }
11 }
12 class Fan implements Switchable
13 {
14     public void turnOn()
15     {
16         System.out.println(x: "Fan is ON");
17    }
18 }
19 public class Main12
20 {
21     Run main | Debug main | Run | Debug
22     public static void main(String[] args)
23     {
24         Switchable s1 = new Light();
25         Switchable s2 = new Fan();
26
27         s1.turnOn();
28         s2.turnOn();
29     }
30 }
```

Output:



The screenshot shows the Visual Studio Code interface with the Output window open. The menu bar at the top includes File, Edit, Selection, View, Go, and Run. The Output window has tabs for PROBLEMS (23), OUTPUT, and DEBUG CONSOLE. The OUTPUT tab is active, displaying a PowerShell terminal session. The terminal shows the current directory as C:\Users\Ayush\projects and lists the contents of the directory: Users\Ayush\AppData\Roaming\Code, Light is ON, Fan is ON, and a file named PS C:\Users\Ayush\projects>.

```
PS C:\Users\Ayush\projects>
PS C:\Users\Ayush\projects>
Users\Ayush\AppData\Roaming\Code
Light is ON
Fan is ON
PS C:\Users\Ayush\projects>
```